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AZ-900 Study Notes

Comprehensive study notes covering all Azure Fundamentals exam domains with detailed explanations, tables, and exam tips.

Microsoft Azure Fundamentals Certification Study Notes

Code: AZ-900

Cloud Concepts (25-30%)

Cloud Computing Fundamentals

Essential cloud computing concepts for AZ-900

> **Domain Weight**
>
> This domain represents 25-30% of the exam. Focus on understanding cloud concepts and Azure's value proposition.

What is Cloud Computing?

- **Definition:** Delivery of computing services (servers, storage, databases, networking, software) over the internet
- **On-Demand:** Provision resources as needed without upfront investment
- **Scalability:** Increase or decrease resources based on demand
- **Elasticity:** Automatically scale resources in response to demand
- **Pay-as-you-go:** Only pay for resources you actually consume

Analytics (⌘⇧A)
Feedback

CapEx vs OpEx

Capital Expenditure (CapEx)	Operational Expenditure (O
Upfront investment in physical infrastructure	Pay for services as you use them
Depreciated over time	Deducted in same billing pe
Requires accurate forecasting	Flexible, adjust as needs change
High initial cost	Lower initial cost
You own the infrastructure	Provider owns infrastructur

> **Consumption-Based Model**
>
> Cloud computing uses a consumption-based model where you only pay for what you use. This provides better cost prediction and eliminates waste from overprovisioned resources.

Cloud Service Models

IaaS, PaaS, and SaaS explained

Service Model Comparison

Model	Description	You Manage	Provider Manages	Azure I
IaaS	Infrastructure as a Service	OS, runtime, apps, data	Hardware, networking, virtualization	VMs, V Storag
PaaS	Platform as a Service	Apps and data only	Everything else including OS	App Se Azure S

Model	Description	You Manage	Provider Manages	Azure I
				Functic
SaaS	Software as a Service	Just use the application	Everything	Micros Dynam

Shared Responsibility Model

Always Your Responsibility

Information and data, devices, accounts and identities

Always Azure Responsibility

Physical datacenter, physical network, physical hosts

Varies by Service Model

Operating system, network controls, applications, identity infrastructure

Exam Tip

Know what you manage vs what Azure manages for each service model. IaaS = most customer responsibility, SaaS = least customer responsibility.

Cloud Deployment Models

Public, Private, and Hybrid Cloud

Deployment Model Comparison

Model	Description	Advantages	Disadvantages
Public Cloud	Services offered over public internet, owned by cloud provider	No CapEx, agility, pay-as-you-go, no maintenance	Less control, security concerns, compliance challenges
Private Cloud	Computing resources used exclusively by one organization	More control, customization, legacy support	CapEx required, limited scalability, maintenance burden
Hybrid Cloud	Combines public and private clouds	Flexibility, compliance options, gradual migration	Complexity, integration challenges, cost management

Hybrid Cloud Use Cases

- **Cloud Bursting:** Use public cloud for peak demand while running baseline on-premises
- **Compliance:** Keep sensitive data on-premises while using cloud for other workloads
- **Gradual Migration:** Move workloads to cloud incrementally
- **Disaster Recovery:** Use cloud as backup site for on-premises infrastructure

Benefits of Cloud Computing

Key advantages of using cloud services

Core Benefits

High Availability

Resources available when needed, typically expressed as percentage uptime (99.9% = 8.76 hours downtime/year)

Scalability

Vertical (scale up/down) - add CPU/RAM. Horizontal (scale out/in) - add/remove instances

Elasticity

Automatically scale resources based on actual demand in real-time

Reliability

Decentralized design enables deployment in multiple regions for fault tolerance

Predictability

Predictable performance through autoscaling and load balancing; predictable costs through tracking and forecasting

Security

Cloud providers offer security tools and features; customers retain control over implementation

Governance

Templates and auditing help ensure resources comply with corporate standards

Manageability

Manage through portal, CLI, APIs, or PowerShell; autoscaling and template-based deployment

SLA (Service Level Agreement)

Azure commits to specific uptime percentages for each service. Higher availability = more complex architecture (multiple regions, availability zones). Know that 99.99% (four 9s) allows only ~52 minutes of downtime per year.

Azure Architecture (35-40%)

Core Architectural Components

Azure regions, availability zones, and resource organization

> **Highest Weighted Domain**
>
> This domain represents 35-40% of the exam. Know Azure architecture components thoroughly.

Azure Global Infrastructure

Component	Description	Key Points
Regions	Geographic area with one or more datacenters	60+ regions worldwide, choose based on compliance, latency, service availability
Region Pairs	Two regions paired for disaster recovery	At least 300 miles apart, automatic replication for some services
Availability Zones	Physically separate datacenters within a region	Minimum 3 zones per enabled region, 99.99% VM SLA
Sovereign Regions	Isolated from main Azure (US Gov, China)	Separate instances for compliance requirements

Resource Organization Hierarchy

- **Management Groups:** Containers for multiple subscriptions, apply governance at scale
- **Subscriptions:** Billing boundary and access control boundary
- **Resource Groups:** Logical container for related resources, cannot be nested
- **Resources:** Individual Azure services (VMs, databases, storage accounts)

> **Key Facts**

- Resources can only exist in ONE resource group
- Resource groups cannot be nested
- Deleting a resource group deletes ALL resources inside it
- Resources in a group can be in different regions

Azure Compute Services

Virtual machines, containers, and serverless options

Compute Options

Service	Type	Use Cases	Key Feature
Virtual Machines	IaaS	Full OS control, lift-and-shift	Windows/L any worklo VM Scale S for auto-sc
App Service	PaaS	Web apps, APIs, mobile backends	Auto-scalin CI/CD, mult languages, infrastructu manageme
Azure Functions	Serverless	Event-driven, microservices	Pay per execution, auto-scale zero, trigge from many sources
Container Instances	Containers	Simple container workloads	Fast startup per-second billing, no orchestratio
Azure Kubernetes Service	Containers	Complex container	Managed Kubernetes

Service	Type	Use Cases	Key Feature
		orchestration	auto-scalin self-healing
Azure Virtual Desktop	VDI	Remote desktop, app virtualization	Multi-sessi Windows 1(Microsoft 3 Apps

VM Availability Options

Availability Sets

Group VMs across fault domains (rack) and update domains (maintenance). Protects from hardware failure and planned maintenance.

Availability Zones

Deploy VMs across physically separate datacenters. Protects from entire datacenter failure. 99.99% SLA.

VM Scale Sets

Deploy and manage identical VMs. Auto-scale based on demand or schedule. Up to 1000 VMs per set.

Azure Networking Services

Virtual networks, connectivity, and content delivery

Core Networking Services

Service	Description	Key Features
Virtual Network (VNet)	Isolated network in Azure	Subnets, NSGs, service endpoints, private endpoints

Service	Description	Key Features
VNet Peering	Connect two VNets	Low latency, private connectivity, cross-region supported
VPN Gateway	Encrypted connection over internet	Site-to-site, point-to-site, VNet-to-VNet
ExpressRoute	Private dedicated connection	Higher bandwidth, lower latency, no internet, more reliable
Azure DNS	Host DNS domains in Azure	Alias records, private DNS zones, anycast hosting
Load Balancer	Distribute traffic	Layer 4, internal or public, zone-redundant
Application Gateway	Web traffic load balancer	Layer 7, SSL termination, WAF, URL-based routing
Azure CDN	Content delivery network	Cache content at edge locations, reduce latency

Network Security

Network Security Groups (NSG)

Filter traffic to/from Azure resources. Rules based on source, destination, port, protocol. Applied to subnet or NIC.

Azure Firewall

Managed firewall service. Stateful, high availability, unrestricted scalability. Threat intelligence filtering.

Azure DDoS Protection

Basic (free, automatic) or Standard (advanced protection with monitoring and mitigation).

Azure Storage Services

Blob, File, Queue, Table, and Disk storage

Storage Account Types

Storage Type	Description	Use Cases
Blob Storage	Object storage for unstructured data	Images, videos, backups, data lakes, streaming
Azure Files	Fully managed file shares	Lift-and-shift, shared application settings, SMB/NFS
Queue Storage	Message storage for async processing	Decouple components, workload scheduling
Table Storage	NoSQL key-value store	Semi-structured data, rapid development
Disk Storage	Block-level storage for VMs	OS disks, data disks for VMs

Blob Storage Tiers

Tier	Access Pattern	Minimum Storage	Access Cost
Hot	Frequently accessed	None	Lowest

Tier	Access Pattern	Minimum Storage	Access Cost
Cool	Infrequently accessed (30+ days)	30 days	Lower storage, higher access
Cold	Rarely accessed (90+ days)	90 days	Even lower storage, higher access
Archive	Rarely accessed (180+ days)	180 days	Lowest storage, highest access, hours to retrieve

Storage Redundancy

LRS (Locally Redundant)

3 copies in single datacenter. Lowest cost, protects from drive failure.

ZRS (Zone Redundant)

3 copies across availability zones. Protects from datacenter failure.

GRS (Geo Redundant)

LRS + async copy to paired region. Protects from regional disaster.

GZRS (Geo-Zone Redundant)

ZRS + async copy to paired region. Highest durability available.

Azure Database Services

Managed database offerings in Azure

Database Options

Service	Type	Best For
Azure SQL Database	Managed SQL Server	Modern cloud apps, predictable performance, auto-tuning
Azure SQL Managed Instance	SQL Server PaaS	Lift-and-shift, near 100% SQL Server compatibility
Azure Cosmos DB	Multi-model NoSQL	Global distribution, multiple APIs (SQL, MongoDB, Cassandra)
Azure Database for MySQL	Managed MySQL	MySQL workloads, high availability, auto-backups
Azure Database for PostgreSQL	Managed PostgreSQL	PostgreSQL workloads, Hyperscale option
Azure Cache for Redis	In-memory cache	Session store, caching, real-time analytics

Cosmos DB APIs

Cosmos DB supports multiple APIs: NoSQL (native), MongoDB, PostgreSQL, Cassandra, Gremlin (graph), and Table. Choose based on your application needs and existing expertise.

Management & Governance (15-20%)

Azure Management Tools

Tools for managing Azure resources

Management Interfaces

Tool	Description	Best For
Azure Portal	Web-based graphical interface	Visual management, learning Azure, one-time tasks
Azure CLI	Cross-platform command line	Scripting, automation, Bash users
Azure PowerShell	PowerShell module	Windows admins, complex scripts, existing PS expertise
Azure Cloud Shell	Browser-based shell	Quick access from anywhere, no local install needed
Azure Mobile App	iOS/Android app	Monitoring, quick actions on the go

Infrastructure as Code

ARM Templates

JSON files that define resources to deploy. Declarative, idempotent, modular.

Bicep

Simpler syntax than ARM JSON. Compiles to ARM templates. Native Azure tooling.

Terraform

Third-party IaC tool. Multi-cloud support. HCL syntax.

Monitoring and Governance

Azure Monitor, Advisor, and governance tools

Monitoring Services

Service	Purpose	Key Features
Azure Monitor	Comprehensive monitoring platform	Metrics, logs, alerts, dashboards, Application Insights
Azure Service Health	Track Azure service status	Service issues, planned maintenance, health advisories
Azure Advisor	Personalized recommendations	Cost, security, reliability, performance, operational excellence
Log Analytics	Query and analyze logs	KQL queries, workbooks, integration with Monitor

Governance Tools

Azure Policy

Enforce organizational standards. Audit or deny non-compliant resources. Built-in and custom policies.

Azure Blueprints

Package of ARM templates, policies, RBAC, resource groups. Repeatable, auditable deployments.

Resource Locks

Prevent accidental deletion or modification. CanNotDelete or

Tags

Metadata for resources. Organize, track costs, automate.

ReadOnly. Inherited by child resources.

Not inherited by default.

Policy vs RBAC

Azure Policy: What resources can be created or configured (resource properties)

RBAC: What users can do with existing resources (user permissions)

Security & Identity (20-25%)

Identity and Access Management

Azure AD, authentication, and authorization

Microsoft Entra ID (Azure AD)

- **Cloud-based identity:** Authentication and authorization service
- **Single Sign-On (SSO):** One set of credentials for multiple applications
- **Application management:** Register and manage app access
- **B2B and B2C:** External identity management options
- **Device management:** Manage device access to resources

Authentication Methods

Multi-Factor Authentication

Something you know (password) + something you have (phone) + something you are (biometric)

Passwordless

Windows Hello, FIDO2 keys, Microsoft Authenticator app

Conditional Access

If-then policies based on signals (user, location, device, app, risk)

Role-Based Access Control (RBAC)

Built-in Role	Permissions	Scope
Owner	Full access + can delegate	Management group, subscription, resource group, resource
Contributor	Full access, cannot delegate	Same as Owner
Reader	View only	Same as Owner
User Access Administrator	Manage user access only	Same as Owner

> **Principle of Least Privilege**
>
> Always grant the minimum permissions needed to perform a task. Use custom roles when built-in roles provide too much access.

Azure Security Services

Microsoft Defender, Sentinel, and security tools

Security Services

Service	Purpose	Key Features
Microsoft Defender for Cloud	Cloud security posture management	Secure score, recommendations, threat protection

Service	Purpose	Key Features
Microsoft Sentinel	Cloud-native SIEM/SOAR	Log aggregation, AI analytics, automated response
Azure Key Vault	Secrets management	Keys, secrets, certificates, HSM-backed
Azure DDoS Protection	DDoS mitigation	Basic (free) or Standard (advanced)
Azure Firewall	Managed network firewall	Stateful, threat intelligence, logging

Zero Trust Model

Verify Explicitly

Always authenticate and authorize based on all available data points

Least Privilege Access

Limit user access with just-in-time and just-enough-access (JIT/JEA)

Assume Breach

Minimize blast radius, segment access, verify end-to-end encryption

Defense in Depth

- **Physical security:** Datacenter access controls
- **Identity & access:** Authentication, RBAC
- **Perimeter:** DDoS protection, firewalls
- **Network:** Network segmentation, NSGs
- **Compute:** Secure VMs, endpoint protection

- **Application:** Secure development practices
- **Data:** Encryption at rest and in transit

Pricing & Support (10-15%)

Azure Pricing

Understanding Azure costs and pricing factors

Factors Affecting Cost

- **Resource type:** Different services have different pricing models
- **Consumption:** Pay-as-you-go based on usage
- **Maintenance:** More complex architectures = higher costs
- **Geography:** Prices vary by region
- **Network traffic:** Inbound usually free, outbound charged
- **Subscription type:** Enterprise, pay-as-you-go, free tier

Cost Management Tools

Tool	Purpose	Key Features
Pricing Calculator	Estimate costs before deployment	Configure resources, compare regions, export estimates
TCO Calculator	Compare on-premises vs Azure	Input current infrastructure, see potential savings
Cost Management	Track and analyze spending	Budgets, alerts, cost analysis, recommendations

Tool	Purpose	Key Features
Azure Advisor	Cost optimization recommendations	Right-size VMs, reserved instances, idle resources

Ways to Reduce Costs

Reserved Instances

1 or 3 year commitment for up to 72% savings on VMs, SQL, Cosmos DB

Azure Hybrid Benefit

Use existing Windows Server or SQL Server licenses

Spot VMs

Use unused capacity at up to 90% discount (can be evicted)

Auto-Shutdown

Schedule VMs to stop during non-business hours

Azure Support Plans

Understanding Azure support options

Support Plan Comparison

Plan	Cost	Technical Support	Response T
Basic	Free	Self-help only	N/A
Developer	$29/month	Email during business hours	8 hours
Standard	$100/month	24/7 email and phone	1 hour

Plan	Cost	Technical Support	Response T
Professional Direct	$1,000/month	24/7 + advisory	1 hour + es
Premier	Custom	24/7 + TAM	15 minutes

All Plans Include

- Azure Advisor recommendations
- Azure Service Health
- Azure documentation and community support
- Billing and subscription support

Azure SLA (Service Level Agreement)

- **99.9% (three 9s):** 8.76 hours downtime per year
- **99.95%:** 4.38 hours downtime per year
- **99.99% (four 9s):** 52.56 minutes downtime per year
- **Composite SLA:** Multiply individual SLAs for dependent services
- **No SLA:** Free tier services and preview features

Azure Service Lifecycle

Preview features and service updates

Service Lifecycle Stages

Stage	Description	SLA	Support
Private Preview	Limited access, invitation only	No	Limited

Stage	Description	SLA	Support
Public Preview	Available to all, may have limitations	No	Limited
General Availability (GA)	Production ready	Yes	Full

> **Preview Considerations**
>
> Preview features may change, be discontinued, or have different pricing at GA. Not recommended for production workloads unless you accept the risks.

Staying Informed

- **Azure Updates:** Official announcements for new features and changes
- **Azure Blog:** In-depth articles and announcements
- **Azure Friday:** Video series with Azure engineers
- **Microsoft Learn:** Free training and documentation

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